Linear Regression Algorithm

$h_\theta(x) : \theta_0 + \theta_1 x$

$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots + \theta_n x_n$

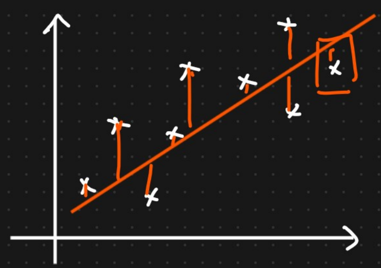
Convergence Algorithm

m = no. of data points

[MSE]

Cost function $J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)} \right)^2$

$h_\theta(x)^{(i)}$ ↓ predicted; $y^{(i)}$ ↓ Truth point



Loss function = $\left(h_\theta(x)^{(i)} - y^{(i)} \right)^2$

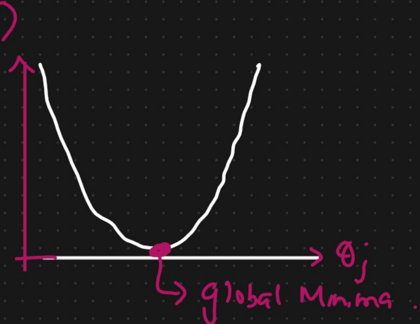
$j = 1$ to m

repeat until convergence

$\{$

$\theta_j := \theta_j - \alpha \boxed{\frac{\partial}{\partial \theta_j} J(\theta_j)}$

$\}$



$\theta_0 \quad \theta_1$

$\frac{\partial}{\partial x}(x)^2 = 2x$

$\frac{\partial}{\partial x}(x)^n = n x^{n-1} (1)$

$\frac{\partial}{\partial x}(x+1)^2 = 2x(x+1) \times (1+0)$

$= 2(x+1)$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_0} \left[\frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)} \right)^2 \right]$$

$h_\theta(x) = \theta_0 + \theta_1 x$

$j = 0$

$$= \frac{\partial}{\partial \theta_0} \left[\frac{1}{m} \sum_{i=1}^{m} \left((\theta_0 + \theta_1 x)^{(i)} - y^{(i)} \right)^2 \right] \Rightarrow \frac{2}{2m} \sum_{i=1}^{m} \left(h_\theta(x) - y^{(i)} \right)$$

$\Downarrow$

$$= \frac{2}{m} \sum_{i=1}^{m} \left[(\theta_0 + \theta_1 x) - y^{(i)} \right] * 1$$

$\boxed{\frac{\partial}{\partial \theta_0} (\theta_0 + \theta_1 x)}$

$(1+0)$

$j=1$

$$= \frac{\partial}{\partial \theta_1}\left[\frac{1}{m}\sum_{i=1}^{m}\left((\theta_0+\theta_1 x)^{(i)} - y^{(i)}\right)^2\right]$$

$$= \frac{2}{m}\sum_{i=1}^{m}\left((\theta_0+\theta_1 x)^{(i)} - y^{(i)}\right) * \left[\ x\ \right]$$

Repeat until Convergence

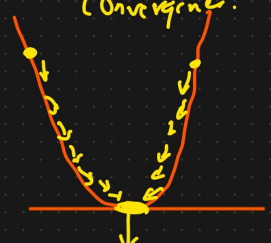
$$\{$$

$$\theta_0 := \theta_0 - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x)^{(i)} - y^{(i)}\right)$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta(x)^{(i)} - y^{(i)}\right) x^{(i)}$$

$$\}$$

α = speed of Convergence.



Cost functions

1. MSE
2. MAE
3. RMSE

Linear Algebra

$(a-b)^2 = \boxed{a^2-2ab+b^2}$ $\Rightarrow$ $\boxed{ax^2+by+c=0}$

$y = mx + c$

1. MSE {Mean Squared Error}

$$MSE = \frac{\sum_{i=1}^{n}(y-\hat{y})^2}{n}$$

↳ quadratic Equation

$\boxed{\theta_0 \ \& \ \theta_1}$

$\hat{y} = \theta_0 + \theta_1 x$

↳ predicted value

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n}\left[\left(y-(\theta_0+\theta_1 x)\right)^2\right]$$

Advantages

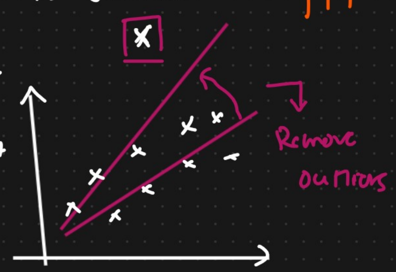
1. This equation is differentiable
2. This equation also has one global minima.

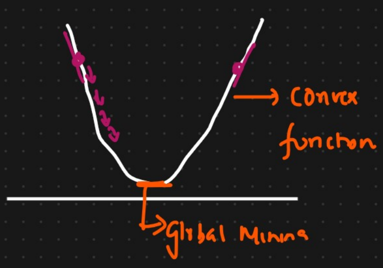
Disadvantage

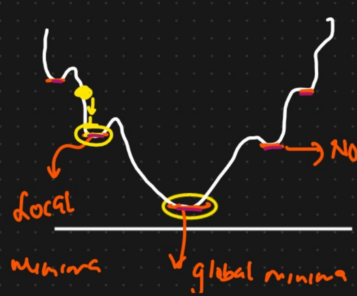
1. This is not robust to outliers
2. Penalizing the Error

Changing the unit

Cost function ↑↑↑







Exp	Salary (y)	
–	–	1 lakh
–	–	
–	–	

← Dependent feature

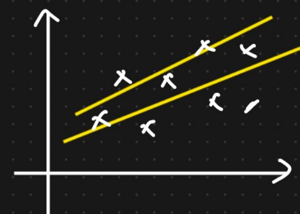
$(y-\hat{y})^2$

$(\text{lakh})^2$

$$MSE = \frac{\sum_{i=1}^{n}(y-\hat{y})^2}{n}$$ ⇒ cost function ↓↓↓↓

$|5-8| = |-3| = 3$

② Mean Absolute Error (MAE)



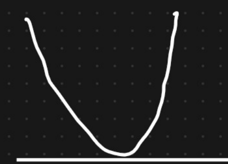
$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y-\hat{y}|$$

Advantage

① Robust to outliers

② It will also be in the same unit.

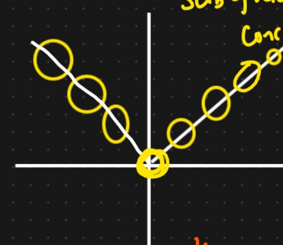
Disadvantage



① Convergence usually takes more time. Optimization is a complex task

② Time Consuming

Sub gradient concept



① Huber loss ⇓ MSE and MAE

② RMSE ⇓ $\sqrt{MSE}$ ⇒ Advantage

Unit ↓ | Outliers ↓ | Differentiable ↓ ⇒ Disadvantages

Performance Metrics

Loss function : MAE, MSE, RMSE, Huber loss

① R squared

② Adjusted R Squared.

① R squared

SS_{Res} = Sum of square Residuals.

SS_{Total} = Sum of square Average.

$$R\ \text{squared} = 1 - \frac{SS_{Res}}{SS_{Total}}$$

R^2 squared be -ve

⇓

Model is very bad

$$= 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \quad \begin{matrix} \rightarrow \text{low value} \\ \rightarrow \text{high value} \end{matrix}$$

$\bar{y}$: Average of y

$$R\ \text{Squared} = 1 - \left\{ \frac{\text{Small number}^{\downarrow \text{Big}}}{\text{Bigger number}_{\downarrow \text{Small}}} \right\} \rightarrow \text{Small number}$$

$\boxed{\leq 1}$

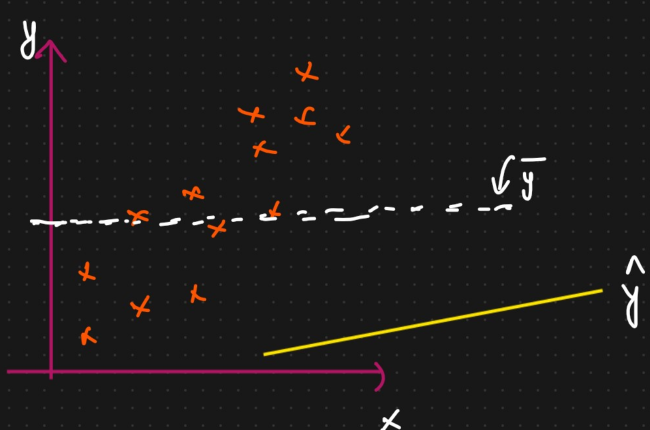
= 0.85

↳ 85% Accurate

= 0.75 → 75% Accurate.

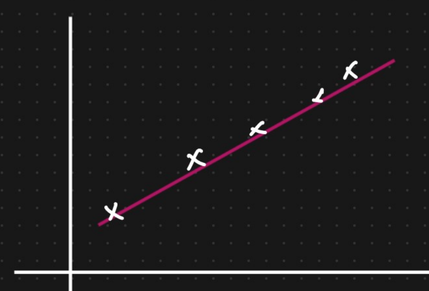
Performance of the Model that you have Created.

R^2 Squared {-ve}



R^2 = -ve

$\boxed{R^2 = 1}$



② Adjusted R squared ✓

{overfitting And underfitting}

Size of House | City Location | No. of bedrooms | Gender | Price

R^2	65% ✓	Adj R^2 = 63%	P=1
R^2	75% ✓	Adj R^2 = 73%	P=2
R^2	88% ✓		
R^2	90%		

⇒ No.

Adjusted R^2 85%

$$\boxed{\text{Adjusted } R^2 = 1 - \frac{(1-R^2)(N-1)}{N-P-1}}$$

N = No. of data points

P = No. of Independent feature.

Overfitting And Underfitting (Bias And Variance).

DATASET

Exp	Year
—	—
—	—
—	—
—	—
—	—
—	—
—	—
—	—
—	—

DATASET

1000 datapoints → TRAINING (700) → Model Training

1000 datapoints → TEST (300) → Model Test

TEST ⇓ New DATASET

700

TRAINING DATASET

500 → TRAIN ↓ TRAIN THE MODEL

200 → VALIDATION ⇓ Hyperparameter Tuning the Model

Model

TRAIN DATA	Very Good Accuracy (90%) [Low Bias]
TEST DATA	Very Good Accuracy (85%) [Low Variance]

↳ Generalized Model

Exp	Age
—	—
—	—
—	—
—	—

TRAIN Very Good Accuracy [90%] [Low Bias]

TEST Bad Accuracy (50%) [High Variance]

⇒ Hyperparameter Tuning

⇓

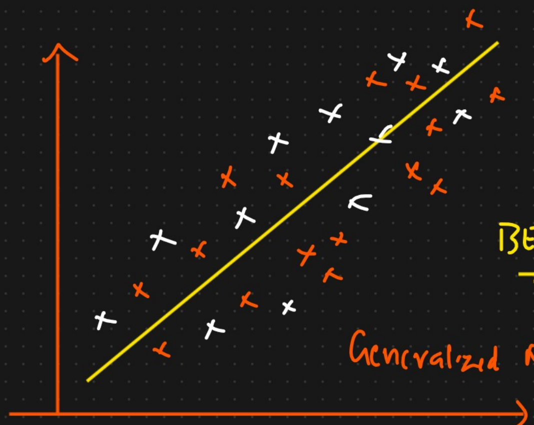
Overfitting

Model Accuracy is Low [High Bias]

Model Accuracy is Low/High [Low or High Variance]

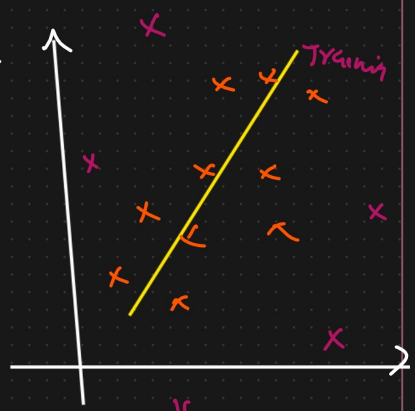
⇓

Model is underfitting



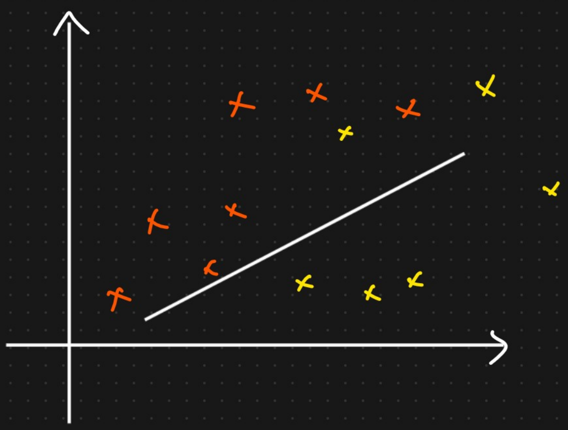
x → TRAINING DATA

BEST FIT



⇓

Overfitting



⇒ Underfitting

[WEEKDAYS]

⇓

Ridge, Lasso Regression

Elastic Regression

SATURDAY ⟶ Linear Regression